STACKS, DERIVED CATEGORIES OF SPACES: PERFECT-DIRECT-IMAGE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5745-5756 of sources/stacks-spaces-perfect/pinned-original-source.tex; proof span is lines 5758-5791, the complete author proof (32 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 35 of queue_spaceperfect.json (pb=1656, stmt 5745-5756, proof 5758-5791).

Editorial note. stacks-spaces-perfect wave sp01 (2026-10-14): fresh adjudication accept per lane-s3/sp01_judgments; pool per Chief 12:4x unfreeze (reserve-pool workflow, six-check clean); spaces-perfect.tex 126 with-proof rows, 70 b107-occupied labels avoided, 56 free rows judged 32 accept / 24 reject.

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-perfect-direct-image' also exists in stacks-perfect — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 08IN: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 4883–4892. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08GK: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1411–1417. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08IK: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 3324–3331. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08IL: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 3369–3378. Cited by the retained statement or proof but

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not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CZH: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1210–1217. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08G8: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 5791–5791. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08I6: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 5791–5791. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let Y be a Noetherian algebraic space over S. Let $f : X \to Y$ be a morphism of algebraic spaces which is locally of finite type and quasi-separated. Let $E \in D(\mathcal{O}_X)$ such that*

1. *$E \in D^b_{Coh}(\mathcal{O}_X)$,*
2. *the support of $H^i(E)$ is proper over Y for all i,*
3. *E has finite tor dimension as an object of $D(f^{-1}\mathcal{O}_Y)$.*

Then Rf_*E is a perfect object of $D(\mathcal{O}_Y)$.

Proof. By Lemma 08GK we see that Rf_*E is an object of $D^b_{Coh}(\mathcal{O}_Y)$. Hence Rf_*E is pseudo-coherent (Lemma 08IK). Hence it suffices to show that Rf_*E has finite tor dimension, see Cohomology on Sites, Lemma 08G8. By Lemma 08IL it suffices to check that $Rf_*(E) \otimes^{\mathbf{L}}_{\mathcal{O}_Y} \mathcal{F}$ has universally bounded cohomology for all quasi-coherent sheaves $\mathcal{F}$ on Y. Bounded from above is clear as $Rf_*(E)$ is bounded from above. Let $T \subset |X|$ be the union of the supports of $H^i(E)$ for all i. Then T is proper over Y by assumptions (1) and (2) and Lemma 0CZH. In particular there exists a quasi-compact open subspace $X' \subset X$ containing T. Setting $f' = f|_{X'}$ we have $Rf_*(E) = Rf'_*(E|_{X'})$ because E restricts to zero on $X \setminus T$. Thus we may replace X by X' and assume f is quasi-compact. We have assumed f is quasi-separated. Thus

$$Rf_*(E) \otimes^{\mathbf{L}}_{\mathcal{O}_Y} \mathcal{F} = Rf_*\left(E \otimes^{\mathbf{L}}_{\mathcal{O}_X} Lf^*\mathcal{F}\right) = Rf_*\left(E \otimes^{\mathbf{L}}_{f^{-1}\mathcal{O}_Y} f^{-1}\mathcal{F}\right)$$

by Lemma 08IN and Cohomology on Sites, Lemma 08I6. By assumption (3) the complex $E \otimes^{\mathbf{L}}_{f^{-1}\mathcal{O}_Y} f^{-1}\mathcal{F}$ has cohomology sheaves in a given finite range, say $[a, b]$. Then Rf_* of it has cohomology in the range $[a, \infty)$ and we win. □